

10 - Group 2

20/12/24

Group 2 elements

→ Be, Mg, Ca, Sr, Ba, Ra

→ also called 'alkaline earth metals' and are very reactive

→ down the group, reactivity ↑

Properties of Group 2 elements

⇒ atomic radius

increases down the group due to increase in electron shells

⇒ melting point

decreases down the group because metallic bonding weakens as atom size increases

⇒ reactivity

increases down the group

⇒ ionization energy

decreases down the group because outer electrons are further from nucleus

⇒ basicity of compounds

Group 2 elements form basic oxides/hydroxides & their basicity increases down the group because of the increased solubility of hydroxides

Reactions with oxygen

$2\text{Mg (s)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{MgO (s)}$; burns with a brilliant white flame

$2\text{Ca (s)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{CaO (s)}$; burns with a brick red flame

$2\text{Sr (s)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{SrO (s)}$; burns with a crimson red flame

$2\text{Ba (s)} + \text{O}_2 \text{ (g)} \rightarrow 2\text{BaO (s)}$; burns with an apple-green flame

Reactions with water

$\text{Mg (s)} + \text{cold water} \rightarrow \text{Mg(OH)}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$ slow reaction

$\text{Mg (s)} + \text{steam} \rightarrow \text{MgO (s)} + \text{H}_2 \text{ (g)}$ rapid reaction

$\text{Ca (s)} + 2\text{H}_2\text{O (l)} \rightarrow \text{Ca(OH)}_2 \text{ (s)} + \text{H}_2 \text{ (g)}$

$\text{Sr (s)} + 2\text{H}_2\text{O (l)} \rightarrow \text{Sr(OH)}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$

$\text{Ba (s)} + 2\text{H}_2\text{O (l)} \rightarrow \text{Ba(OH)}_2 \text{ (aq)} + \text{H}_2 \text{ (g)}$

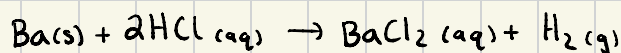
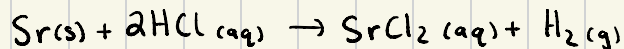
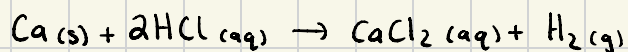
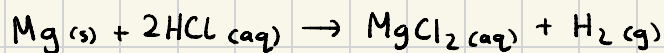
vigorous reaction with cold water

⇒ Ca(OH)_2 appears as a white ppt, just soluble enough to give a weakly alkaline solution

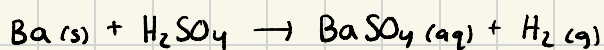
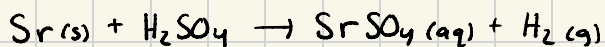
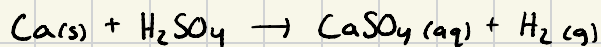
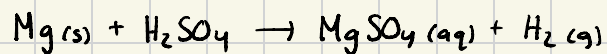
Down the group, the solubility of the metals ↑

Reactions with acids (neutralization)

with HCl

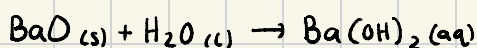
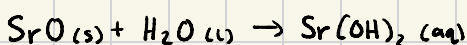
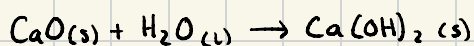
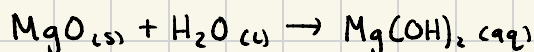


with H₂SO₄



↓ Solubility ↓ down the group

Oxide reactions with water



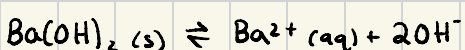
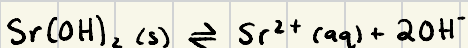
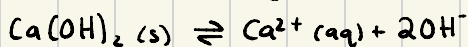
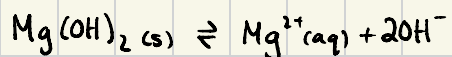
↓ solutions formed from the reactions get more alkaline going down the group

⇒ all group 2 OXIDES are basic. They react with an acid & give the corresponding salt and water

Hydroxide reactions with water

* group 2 hydroxides are not very soluble & don't react with water

* solubility ↑ down the group



↓ dissociate in water & OH⁻ ions make solution more alkaline

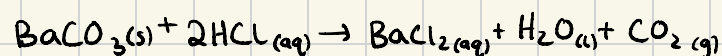
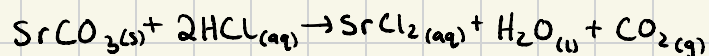
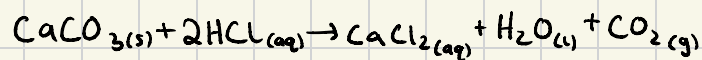
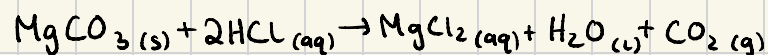
⇒ all group 2 HYDROXIDES are basic. They react with an acid & give the corresponding salt and water

★ down the group, solubility increases for hydroxides & decreases for sulfates ★

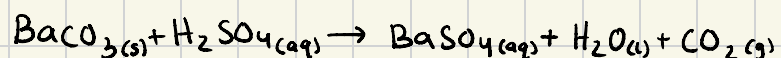
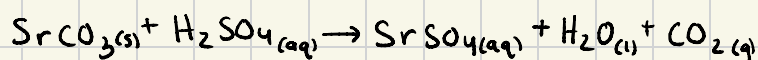
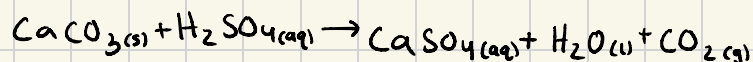
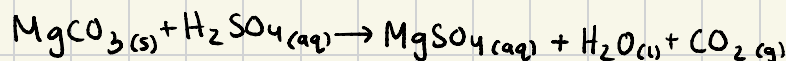
* all group 2 carbonates are insoluble in water

Carbonate reactions with acids

with HCl



with H₂SO₄



Thermal decomposition

* the breakdown of a compound into simpler substances using heat

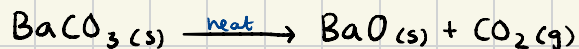
* thermal stability (ability to resist thermal decomposition) ↑ down the group

* more heat is required for the decomposition down the group

* Group 2 *

Thermal decomposition of carbonates

⇒ metal carbonates break down to form the metal oxide & CO₂ gas



Thermal decomposition of nitrates

⇒ metal nitrates break down to form the metal oxide, toxic NO₂ gas & O₂ gas

